Practical 3. Foundations of data and measurement in neuroscience

JXH-1104 Neuroscience of Sport Performance

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Introduction

The lecture introduced the principles of measurement, the description of distributions, and the accurate presentation of data. In this practical, these principles are applied to the data collected by the class in Weeks 1 and 2.

The analysis is carried out in Excel for the web. No prior experience of spreadsheets is assumed: the workbook is prepared in advance, and entries are made only in the cells shaded **yellow**, each confirmed by pressing **Enter**.

Aim

To locate an individual observation within the class distribution, to compute measures of central tendency and dispersion, to examine within-participant difference scores, and to evaluate the accuracy of two graphical presentations of the same data.

Materials

- A computer (e.g., computer room)
- A partner, with whom the tasks are completed
- class workbook (instructions below)

TASK 0: Get a personal copy of the workbook

1. In a web browser, open Excel for the Web and sign in with your Bangor University account.
2. Navigate to blackboard, log in with your Bangor University account, and open this module webpage.
3. Find this document under the Content tab.
4. Find the class workbook and click on it to open it.
5. Select **File → Create a copy → Create a copy online**, retain the suggested name, and select **Save**. The personal copy opens automatically.

Note: The copy is stored in each user's own OneDrive; changes made to it do not affect any other copy.

6. Remain in the browser throughout. Do *not* use the option Open in desktop app.

Alternative procedure

If the option "Save a copy" is unavailable, select **File → Save As → Download a copy**. Then, at **office.com**, select **Upload**, choose the downloaded file, and open it. If neither procedure succeeds, Tasks 2 to 5 can be completed by inspecting the class workbook without editing it, working with a partner on a single copy.

The workbook contains six sheets, listed at the bottom of the screen: **Start here**, **Class data**, **1**, **2**, **3** and **4**. Read **Start here** first, then complete sheets 1 to 4 in order.

TASK 1 / Sheet 1: an individual observation within the class distribution

1. In the yellow cell, enter your Week 1 **block 1 mean reaction time** (ms), i.e. the average reported after the first round of five trials. If this value is not available, enter **268**.
2. The corresponding bar of the histogram is highlighted in red.
3. Note the values reported in the grey cells.

Questions. Where does the observation lie relative to the centre of the distribution? Is the distribution approximately symmetric, or does it have a longer tail on one side?

TASK 2 / Sheet 2: central tendency and dispersion

1. In the first yellow cell, enter **=AVERAGE(times)** and press **Enter**. The adjacent Check cell should read *correct*.
2. In the second yellow cell, enter **=MEDIAN(times)**.
3. In the third yellow cell, enter **=STDEV.S(times)**.
4. Replace any value in column B with **900** and press **Enter**. Observe the three yellow cells.
5. Press **Ctrl+Z** to restore the original value.

Questions. Which of the three statistics changed most when a single extreme value was introduced, and which changed least? Explain the difference with reference to how each statistic is calculated.

TASK 3 / Sheet 3: within-participant difference scores

1. Column D contains, for each participant, the block 2 mean minus the block 1 mean.
2. In the yellow cell, enter **=AVERAGE(diffs)** and press **Enter**.
3. Examine the scatterplot. Each point represents one participant; the dashed line represents identical scores in both blocks.

Questions. Did reaction times change between blocks, on average? How many participants were faster in block 2, and how many slower? To what extent do the points cluster around the line, and what does this indicate about the test–retest reliability of the task?

TASK 4 / Sheet 4: accuracy of graphical presentation

Charts A and B display the same two values from the Week 2 data.

Questions. In which chart does one condition appear considerably slower than the other condition? Compare the starting point of each y-axis. Which chart represents the difference more accurately, and why?

Points for discussion

- The effect of a single extreme observation on the mean and on the median.
- Whether a difference of 20 ms between a participant's two blocks reflects unreliability of the measure or genuine variation in performance.
- The magnitude of the difference between the auditory and visual conditions in Week 2, and which chart conveys it accurately.

Additional exercises

- On sheet 4, select the chain labels and the two columns of values (hold **Ctrl** to select the second range), then select **Insert → Chart → Column**. Add a title. Confirm whether the y-axis begins at zero.
- On the **Class data** sheet, examine the **Device** column. Do smartphones and laptops appear to differ? Would such a difference constitute random or systematic error?
- Using the mean and standard deviation from sheet 2, calculate mean − SD and mean + SD. Using the histogram on sheet 1, estimate the proportion of the class falling between these two values.

If the workbook cannot be opened

Consider the following ten reaction times (ms): 245, 262, 271, 278, 284, 290, 297, 305, 318, 900. Calculate the mean and the median by hand; then remove the value of 900 and calculate both again.